

Clinical NLP for Information Extraction from Electronic Health Records

Kristina Palesheva
MSc in Big data

Transforming unstructured
clinical notes into structured
morbidity data to support
faster, safer decision-making



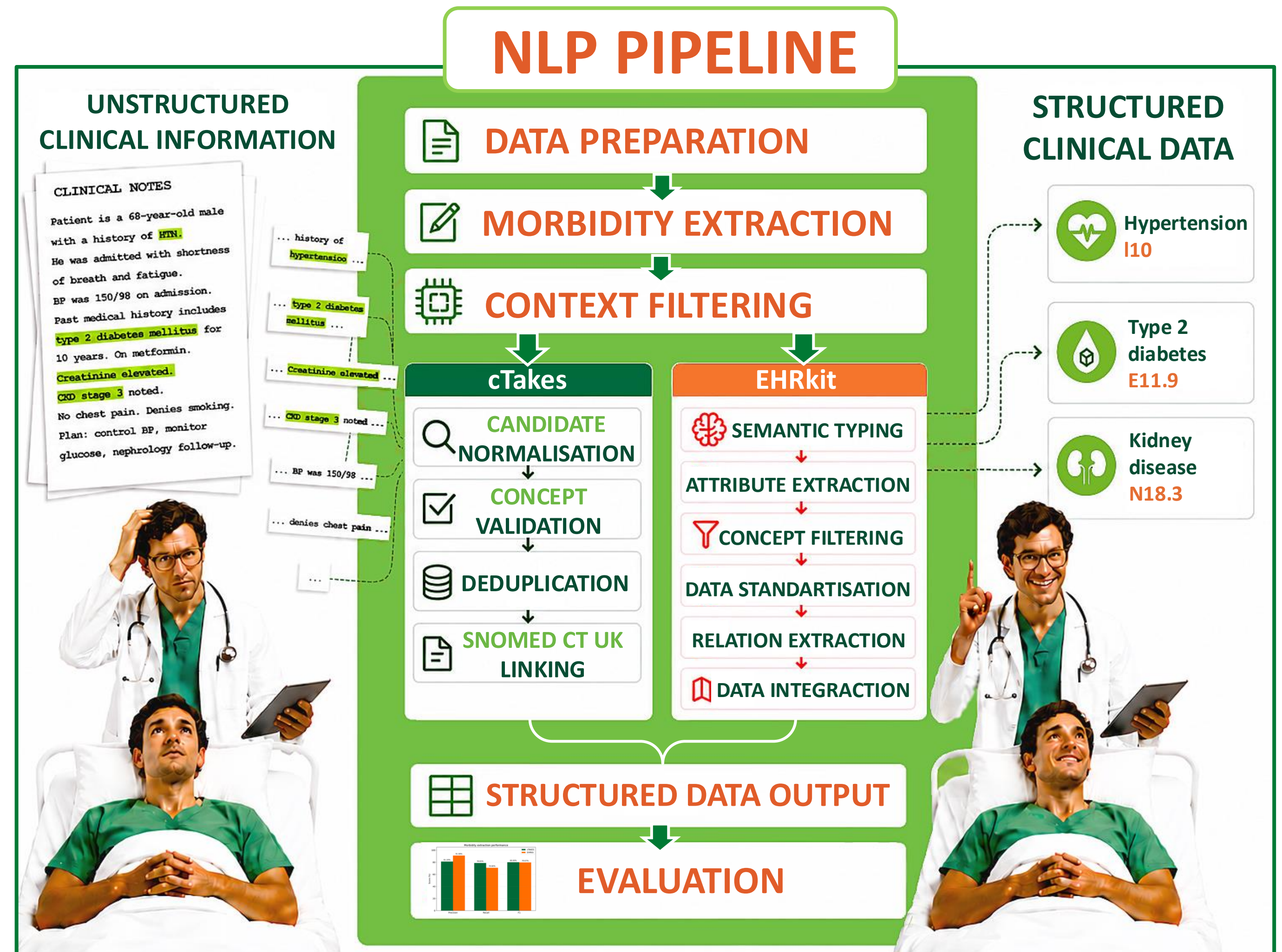
WHY IT MATTERS

Important clinical information is often embedded in long free-text medical notes. **Reliable NLP tools can convert this text into structured data**, making relevant info easier to identify, analyse and reuse.



WHAT WAS DONE

- **Bio_ClinicalBERT** was trained within the EHRKit environment for morbidity extraction.
- **EHRKit** and **cTAKES** were applied to the same unseen clinical records.
- Official NHS **ICD-10 codes family** were assigned to the extracted morbidities.



RESULTS EHRKit achieved higher precision than cTAKES (91.08% vs 81.20%), but cTAKES achieved higher recall (78.83% vs 70.80%) and a slightly higher F1 score (80% vs 79.67%). cTAKES also provided 95.83% accuracy, compared with 94.29% accuracy for EHRKit.

